Escape the Evil Asylum Report

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1 INTRODUCTION

This report is designed to go in-depth into the design process, development process, and theoretical topics surrounding the creation of “Escape the Evil Asylum”. This report will cover the theoretical aspects considered during game design, the methods used in development, an explanation of how testing was done, our design inspirations and so much more. I will not waste time, as I anticipate this report will be very lengthy. Thank you for your patience if you do in fact read this entire thing.

2 THE LAYERED TETRAD

The Layered Tetrad, as described by Gibson Bond (2022), extends Jesse Schelle’s original Elemental Tetrad by taking game analysis and dividing it into 3 layers. These layers are the Inscribed Layer (the game as it is without any player interaction, the rules that the developer implicitly creates), the Dynamic Layer (the game as it is played by people and actually experienced, with all of the individual systems working together), and the Cultural Layer (the layer which reflects the game’s relationship to society and its culture). Each of these layers looks at the 4 tetrad elements of Mechanics, Story, Aesthetics, and Technology, all at different levels of abstraction from each other. These layers, when combined, provide a complete framework for both creating and designing a game, as well as reviewing, experiencing, and critiquing a game. As stated by Schell (2019), no single part of the tetrad can be fully understood by itself. Each element supports and builds on the others. A successful game will achieve a sort of balance across all 4 aspects. This first section of my report applies the Layered Tetrad to “Escape the Evil Asylum”, which is broadly defined as a first-person survival horror game where the player must navigate and escape the asylum they’re trapped in by finding items and solving puzzles, all while avoid and evade a ghost haunting an asylum while they do it.

2.1 Inscribed Layer

The Inscribed Layer of the game described the game as a single, fixed object. The systems encoded by the developer, the map, etc, all as it exists statically. This exists regardless of whether or not anyone is playing it. Gibson Bond (2022) says the inscribed layer represents the intentions of a designer. Scheil (2019) described it as the “game at rest”, and the crystallized form of the designer’s decisions prior to anyone actually playing the game.

2.2 Mechanics (Inscribed)

At the inscribed level, the mechanics of Escape the Evil Asylum consists of a few very clearly defined systems that exist in the game’s code. The core mechanics are picking up items, interacting with levers & doors, and evading the evil ghost haunting the asylum. They must do this while keeping in mind that the ghost will hear if you drop an item, and you can only hold one at a time. So if you open a lock with a key, then come across another key that you will need, you will have to either drop the key you are currently holding to pick up a new one, altering the enemy to your location, or wait until a better opportunity to grab the new key.

The movement and sound detection system is one of the more interesting mechanics of the game in my opinion. The player walks around the prison in a first-person view, trying to avoid being seen by the ghost. While normally, the enemy will follow a fixed path patrolling the room you are currently in, it will break from this if it hears an item that has been dropped. This means that the evasion of the enemy is not a simple fixed “go here after he goes there”, but a dynamic system that changes based on player choices. Item dropping becomes a pivotal mechanic. Players are forced to drop items, as they can only hold one at a time, and more than one item is needed to beat the game. This creates tension that will be elaborated on in a later section, but is worth pointing out here

Another very important inscribed mechanic is the clue system. Since every time you play the game the puzzles are randomized, it is not as simple as memorizing the pattern. You will need to find clues scattered throughout the map that will tell you what levers you need to pull in order to open the door to the next area. Clues are divided into two categories at the design level, red clues and blue clues. The clues written in red are used to mislead the player into thinking they can escape when they really are being lured into a trap, and the blue clues are hints towards the game’s secret ending which has the player banishing the ghost and actually being able to escape at the end. The ghost’s

detection and kill system is also inscribed. The patrol logic and player detection logic is inscribed explicitly in the enemy's AI script. It defines the angle that the enemy can see the player in, the range that they have on their vision, how fast they are during a chase, etc. There is perhaps more to say but that is the core essentials of the system.

2.3 Story (Inscribed)

The narrative of the game is not super complicated, but at first will be a bit tricky to understand. The player starts the game with a screen of text (Would have been an animation, but ran out of time) explaining that they were wandering around the woods when they came across an abandoned asylum. Wanting to explore, they entered a strange looking door. But when they entered, the door disappeared behind them, cutting off their escape. After hearing otherworldly cries down the halls, they come to the conclusion that they are not alone. Then gameplay starts. The game starts with the player walking around a smaller room, with only a single key and a note. This note (written in red) explains that the monster will not let them leave, and that they have to escape. It also says that there are keys and puzzles that need to be solved in order to get out. The player will then likely spend their time finding these keys, finding clues to puzzles (also written in red), and solving them to progress. Eventually they will come across an altar, but not know what it is for. They may also find a note with blue text saying "The monster will not let you escape, but he is not invincible, follow my instructions". But on a first go, most players will likely not know what this could mean and simply move on. However, after opening the third door and entering it, it swiftly shuts behind them. In that final room is a door with a black lock (and no black key exists for the player to find throughout the building), and a single note on the floor (also written in red) saying "Thanks for playing with me. Goodbye". After picking it up, the player loses control of their character, and a death cutscene plays. This reveals to the player that the notes written in red were actually written by the ghost, and were all just part of a game he was playing with no intention of actually letting you leave. Then the Game Over screen will then say "THE END (or is it?)", hinting that there is more. If the player goes back and finds the blue note, they may realise there is more to do. They must find 2 more notes, these ones much more hidden, that are instead written in blue. If you follow the instructions on these notes, the player is alerted to go back to the first room, where the last puzzle is located. Upon completing it, a false wall comes down. If the player then finds the final key to the secret drawer behind this false wall, they find what appears to be a voodoo doll of the ghost enemy, with a note in blue saying "complete the ritual". The player will remember the altar that they saw earlier, and take the doll to the altar. After using it, the ghost will then become trapped in the ritual circle. They will be banished, and a black key will slowly rise from the floor in their place, showing that the monster had the final key the whole time. This then prompts the good ending screen, as the payer has successfully escaped.

2.4 Aesthetics (Inscribed)

The inscribed aesthetics are all of the visual and audio elements that go into making the game. The enemy model, animations, sound effects, etc, all go into making this game what it is on an aesthetic level. As Schell (2019) notes, the aesthetic layer is the most easily and immediately perceived part of the tetrad. Ermi and Mayra (2005) specifically three gameplay components of immersion, those being sensory, challenge-based, and imaginative, all of which Escape the Evil Asylum tries to create. The aesthetic elements shape the emotional experience of the player much more than any other layer, and so they are very important.

Escape the Evil Asylum displays a horror aesthetic built around a confined space and tension. The asylum is meant to feel oppressive and difficult. Darkness is thick, and you cannot see very far. The eerie music that plays in the background is also meant to evoke a sense of dread and tension in the player. As Smith (2023) notes, the pacing and sound-design of a game are some of the most important tools for developers to create tension in the player throughout the gameplay experience. When the player gets in a chase and has to run from and escape the ghost, they see a light getting brighter and brighter behind them as the ghost gets closer and closer (a diegetic danger clue), which evokes tension. After they escape, they can go back to looking for items and clues, solving puzzles, exploring, etc, but will always be wary of the ghost getting them while their back is turned. The red writing of the ghost is also meant to represent danger while the blue writing of a helpful stranger is meant to instill a sense of calm and hope, as they are actually being helpful. The creaking noises that play whenever a drawer opens or a lever is pulled is also meant to make the player uneasy. Did the ghost hear that? The stone door opening noise also makes the player feel like these doors are massive immovable objects, that they have no hope of opening without the proper items.

2.5 Technology (inscribed)

At the inscribed level, technology includes the engine, the platform, that architecture of the code and systems, etc. Escape the Evil Asylum was developed using Unity, targeting Windows (as the final game is an exe). Unity was chosen as it has very easy to implement physics, scripting, and audio while also offering a 3d environment with a built-in enemy navigation mesh. The enemy AI, for example, is programmed largely with Unity's built-in navigation mesh system in mind. It makes pathfinding to the player very easy, with simply in-built commands for setting the enemy's target location to simply update to the player's current location whenever they are seen, or updating their target location to the coordinates of an object when it hits the ground and makes a noise.

2.6 Dynamic Layer

The Dynamic Layer is what emerges from the inscribed mechanics when a player actually plays the game. It describes the complex systems that are created through actual interaction with the systems as they are defined. For example,take two systems. One where an enemy will follow you when they see you, and another where an enemy will go to any sources of noise the player makes. These systems, if put into separate sections of a game, separate enemies, etc, will be experienced completely differently by a player than if they were combined into the same enemy and used at the same time. The emergence of dynamic mechanics is why two games that on paper have similar mechanics might play out very differently, and give players very unique experiences.

2.7 Predicted Emergent Behaviors

All four qualities of the tetrad will be discussed interchangeably, since it is difficult to separate them when they are interacting with each other (hence the term "dynamic"). Since this game has not yet been released to the public and current testing is extremely limited, it can be somewhat difficult to predict what dynamic behaviors would emerge as players played the game. However, from the limited testing and educated guessing based on both developer intention as well as observing how games with similar mechanics work, I can draw some educated guesses. Firstly, it is predicted that the player will, at a certain point, associate the red clues with the enemy, and blue clues with an outside helper whose advice can actually save them. While this is made relatively clear at the end of the bad ending, it is never made explicitly clear. It is simply an intended outcome, and why certain notes are written in red (the enemy is red, glows red, and red is generally associated with danger), while the helpful notes are in blue (calming, peaceful, helpful color). It is hoped that after players realise what is going on, they will replay the game to find the secret, true ending. Secondly, It is predicted that the combination of eerie music, enemy patrolling, chases where the player attempts to evade the enemy, the item dropping mechanic, and the fact that the enemy's actions can

sometimes be difficult to predict, will create a very tense atmosphere for the game. This is intentional of course, as it is meant to be a horror game. Third, it is predicted that the player will eventually come to an important conclusion for how to beat the game more easily. If the player drops an item, the monster will come towards the source of the noise. This is normally bad for the player, as the monster will go towards their location. However, through observation, players will realise that they can actually use this mechanic to manipulate where the monster is going to, giving themselves opportunities to search areas while the monster is at a known position, instead of being somewhere unknown wandering around the building. This was realised both by myself as I developed the game, as well as the first playtester who played the game in its final state. It took roughly 15 minutes and then dying a few times for them to realise this and then start to use it to their advantage. They also started to see the pattern in the ghost's patrol route, and use these things together to create a very optimized playstyle. This shows that while the game may feel difficult to the player at first, they can feel a sense of mastery and accomplishment after figuring out this system and finding a way to use it to their advantage, which creates a very enjoyable player experience.

2.8 Cultural Layer

The cultural layer looks at the relationship between a game, the player playing it, and the effect this has on culture in general. This applies to the game's audience insofar as that may be considered a "culture", and also the general world's reaction to it even if they do not engage directly with the game. Gibson Bond (2022) says that games are not created in a cultural vacuum, and that every mechanic, visual, and narrative decision carries cultural meaning that players will bring their own understanding to, which will give it a new context. I will briefly outline the influence that Escape the Evil Asylum may have on the culture, and the influence culture may have had on the game.

2.9 Mechanics (Cultural)

The mechanics of this game were heavily impacted by the culture in which it exists. They take inspiration from the myriad of horror games that have come before, some of which I have played, others I have only seen or heard about. An example of this is the game Granny, where a player must escape the evil grandma's house while worrying about collecting the items needed to escape. In that game, dropping items creates a noise that

the enemy will hear and go towards. That is where the inspiration for the same mechanic in this game came from. Additionally, other games like slenderman inspired the dark atmosphere of the game. It can be difficult to say where the inspiration for any particular aspect truly came from, as much of the thoughts behind what we make happen on a subconscious level. But these two examples come to mind. The mechanics of being chased, escaping, hiding, and looking for items to escape may cause players to associate it with other games they have played before, such as the games I mentioned earlier, or even other games such as Resident Evil. Things like the single-item inventory limit also will give the game a sense that you can never be prepared for every scenario and that you will always need a choice about what to take and what to leave behind, and this mechanic will certainly affect how the game is perceived culturally in the sense that it will be seen as more difficult than if you could carry multiple items at once.

2.10 Story (Cultural)

The story of this game relies on players knowing what an asylum is/was. In the past (and currently, in other parts of the world), asylums are areas where those who are mentally unfit for society would be put. They were known to be treated very poorly, and be given little consideration. Medical studies with dubious moral considerations would also be performed on these people, as very few would ever notice or care if something went wrong with them. The background knowledge for a place like this informs the player's ideas of what went on here. Who is the ghost? Why are they haunting this place? Were they a former resident of this asylum? Ideas like this will cause the community to create a story in their own minds that is not explicitly written anywhere, but can be inferred based on cultural context. The false hope offered by the ghost's "clues" that eventually leads to the player's demise also shows a narrative of betrayal and trickery. Once may wonder if the ghost experienced something similar in life to end up this way. Again, all of this is based on the player's context for what is going on from the culture they know if in real life, not explicitly written. This allows the story to happen largely without directly being written, letting players fill in the gaps with their own imaginations.

2.11 Aesthetics (Cultural)

The aesthetics of this game are again largely derived from pre-existing cultural elements. The asylum is creepy because people know what asylums were like. The ghost is scary because we know what a ghost is. You likely cannot kill it, and it will probably not be friendly to you. The red color of the ghost, the red glow it gives off (especially during chases as you can see the red glow get progressively brighter as the ghost gets closer and closer to you), and the red writing of the text, is all meant to play

off of our cultural understanding. Red is blood, red is a STOP sign Red is urgency. Blue notes and clues are calm. Blue is the sky. Blue is relaxing. “All will be well”.

However, these associations may not be a universal thing. As Khunnasut (2024) notes, cultural context heavily informs how we interpret visual and aesthetic elements of horror games. In some Asian cultures, red may be seen as representing good fortune and celebration rather than urgency and terror. It must be noted that a player from a different cultural background may not have the same automatic interpretations of the aesthetics that are intended.

2.12 Technology (Cultural)

The technology used in this game is heavily informed by the culture we are in. Unity is a popular game engine, which is free and easy to use. Players are also very familiar with Unity, and have a certain understanding of how Unity games generally play and feel, as well as how items interact with the built–in physics systems, nav-mesh systems, etc. It is important to keep in mind that Escape the Evil Asylum will likely be viewed and compared against other Unity games of the same genre, as it exists in the cultural context of those products. Unity is very well-known, with many tutorials and guides on how to code and script particular things. Many of these were used in the development of this game. It is possible that an observant player may recognize some design elements that were used from other systems, and perhaps even be aware of some exploit to that system that I am not even aware of as the developer. Something to keep in mind as I continue game development even after this project is finished.

2.13 - Layered Tetrad Conclusion

The game’s influence on culture may be comparable in kind (although certainly not in scale) to horror games like Granny or Five Nights at Freddy’s. Containing a sort of secret story and not having a narrative that is made explicitly clear at any point, players may theorize about what events transpired to cause the situation the player is in. Who is the ghost? Why are they here? What happened to them? Who wrote the secret blue clues? Would players compare it to other similar games and like or dislike it? Would the game be intriguing enough for players to want more? All of these things would happen on a cultural level through many different people, so it is difficult (if not impossible) to predict what exactly would happen. But nevertheless, it is important to be mindful of

things that have come before, so we can properly design an experience that players will enjoy and appreciate.

3 DESIGN GOALS

The design goals for this game were as follows: Create a scary game where the player must navigate a building while avoiding a monster which is constantly searching for them. Include items the player must find as well as puzzles they must solve in order to beat the game. Have at least two endings to the game, as well as a unique story for each one. Keep tension throughout the game in order to keep the player engaged. Make the game reasonably difficult, likely requiring at least a few attempts to win, but do not make it frustrating or punishingly difficult. These are the main goals for the design of the game.

The six designer-centric goals are challenge, curiosity, fantasy, fellowship, expression, and submission. The three that will be discussed here are challenge, curiosity, and submission.

The challenge aspect was one of my favorite parts to work on. Fine tuning the game so that it was difficult enough to excite the player, but not too difficult as to devolve into an experience of frustration. One of the ways this was done was with the enemy's speed. When the enemy has a direct line of sight to the player, they move faster than the player does, making escape impossible. The player will realise this as they will either see the enemy gaining on them in a chase, or the enemy's red glow getting brighter behind them as the gap closes. However, if the player turns a corner, goes behind a wall, etc, the enemy will drop in speed by a bit to now be slower than the player. This is not explicitly stated, which makes the player feel challenged by giving them the sense that a chase is inescapable, only to feel pride in their escape if they manage to smartly evade capture. The enemy also goes towards sounds, making the player have to calculate when and where to drop items. Again, a challenge to be figured out. Once the player realises they can drop items to lure the enemy, they gain control over the situation, having mastered a mechanical challenge. The randomized puzzle section also makes it more challenging, as players cannot simply memorize which levers to pull. They must find the clues every single playthrough (or get very very lucky with guessing). One thing I wish there was time to implement was an extreme mode, where the ghost was significantly faster, making the time and location of item drops, as well as keeping track of where the enemy is patrolling, even more crucial.

The second aspect is curiosity. The notes given by both the enemy (red) and the helpful stranger (blue) give the player a sense of curiosity. Who is writing these notes? Are the blue notes from the same person? Where are the items and notes I need in order to win? All of these aspects evoke a sense of curiosity from the player in order to elevate their experience from one of "go here and do this" to one where they get to think about

the experience on a deeper level, and appreciate it in a new way that isn't just mechanical. The player must also be inquisitive enough to find the two secret notes required to get the secret ending, which are hidden much more cleverly than the notes required for the default "bad ending". There is also a general curiosity about the story of the game, as it is not made explicitly clear why the player gets trapped in the first place, why they are being hunted, who the ghost is, etc. This allows the player's imagination to run wild, and create a narrative that is unique to them and their interaction with the game.

The third element is submission. This is the practice of designing systems that players will interact with over and over again, and rules that the player must follow in order to win. This game will generally require more than one playthrough to beat, as the player is likely to make a mistake and get caught on the first few attempts. They must find jets to open doors, find clues to solve puzzles, learn how the monster will run to a dropped item, etc. Players will eventually get familiar with item locations, enemy movement patterns, etc, until they are able to interact with the game in a way where they are able to be successful. The player is taught to master the game through submission to its mechanics and following the rules set out by the developer. If they do this, they will learn how to succeed and successfully evade danger.

The eight player-centric goals, again from Schell's work, are narrative/drama, discovery, challenge, sensation, fantasy, fellowship/social interaction, expression/creativity, and completion. Of these 8, I will focus on the drama, discovery, challenge, and sensation.

In the game, drama occurs as the player attempts to escape. They want to leave the building, escape the ghost, and survive the very tense asylum experience. However, drama also occurs when the player is reading the notes. At a point, they may begin to question who is writing these notes. Whether that be when they find the blue note hinting at another ending, or after they get the bad ending with the red note, realising in shock that they have been deceived by the ghost. This creates a sense of narrative engagement that would not otherwise exist, with the player simply beating the game once and shutting it off, not thinking about it anymore. The intended player experience is for them intrigued about the story, and kept in a sense of tension throughout.

Another playercentric goal is discovery. Players want to discover and find cool things. In Escape the Evil Asylum, players can find secret notes (which are somewhat difficult to find), which will make them feel clever and like they are gaining insight into this

world through their careful and thorough exploration. Their discovery of the secret notes leads them to look for what they are pointing towards, and eventually they will discover the secret ending. The existence of secret story elements and a secret ending will give players who love exploring a greater sense of satisfaction, as their exploration has brought them to a unique part of the game that is not immediately obvious.

Challenge is another player'centric goal. Many players love a challenge. If something is too easy for them, it may be considered a waste of time. As I endeavored to make the game challenging, I do hope that players consider it challenging as well. The speed of the ghost, the fact that it can hear you drop items and open some doors, all of it is in service to creating a more challenging (and tense) game experience. In the future, I wish to add additional difficulty levels to make the game that much harder for people who want to go above and beyond, and are searching for mechanical difficulty.

The final player-centric aspect I will discuss is the sensation I wished to create within the players. The main sensations are fear and tension. By designing the aesthetics in the way they were done, as well as the mechanics being somewhat tricky, I hope to have succeeded in making the game give the players a sense of tension as they try to evade the ghost and survive, as well as fear from the environment as well as the music, sound effects, and knowledge that they are being hunted.

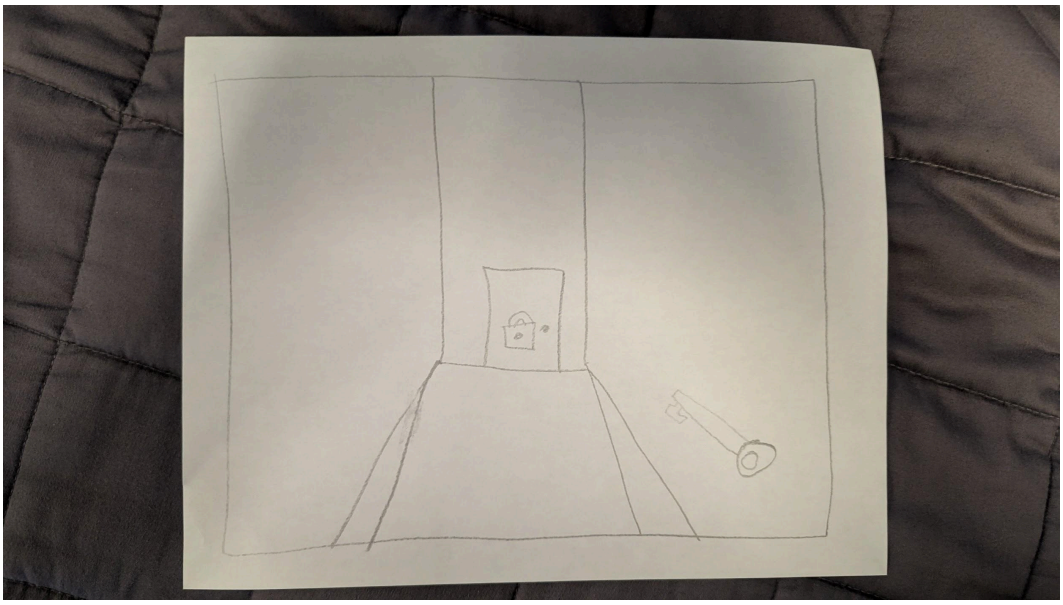
All in all, I feel that the development goals were met very well, and the game I have created has fulfilled them more than adequately. The game is tense, challenging, filled with puzzles and things to keep track of, and has a dynamic story that ends in one of two ways depending on player knowledge and choices.

4 PAPER PROTOTYPING

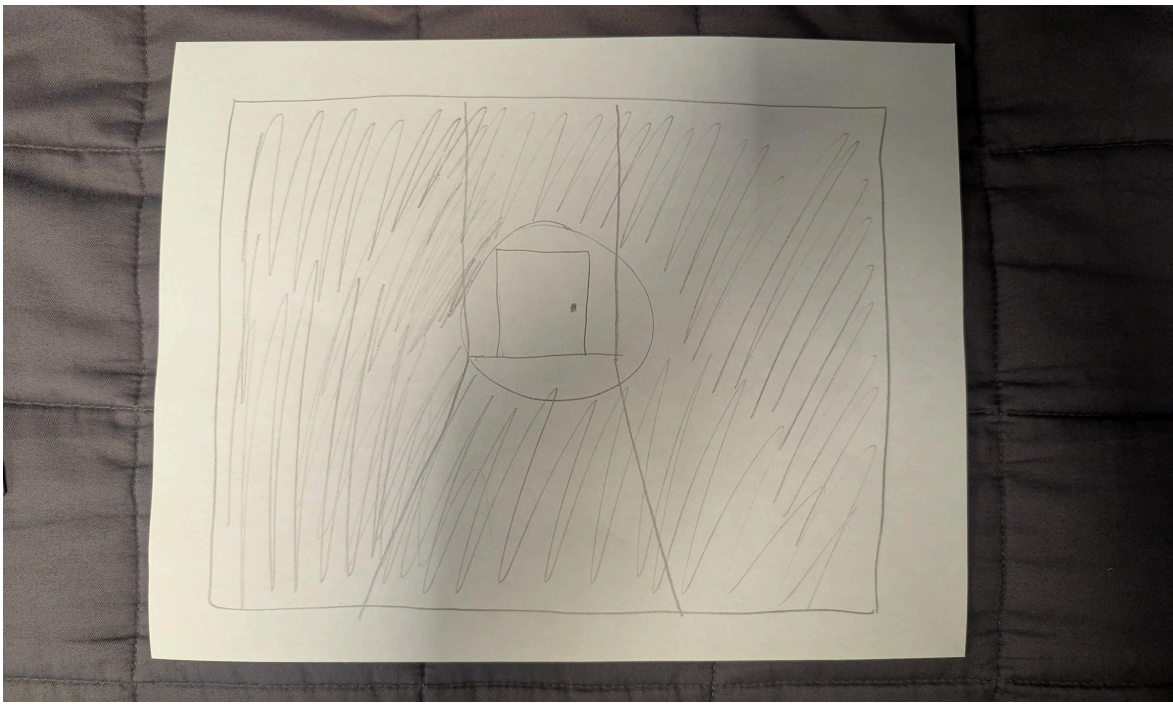
Paper prototyping is generally meant to mean various types of low-fidelity prototypes that can act as the basis of how a game is designed. Be it a menu layout, potential map designs, where enemies will be placed, etc. All of these things can be represented with things like drawings for maps, small cardboard pieces to represent enemies, pennies to represent points, whatever. For this game, we used three primary

things for paper prototyping. First, we drew up some map designs for what the different room layouts could look like. Second, we drew sketches of the enemy to show what the ghost might look like once given a proper 3d model to be used in Unity. Lastly, we created mock-up drawings of things like the main menu, the player's first-person perspective (as in their area of vision with a flashlight in the dark). And a player holding an item. Examples of this are given here:

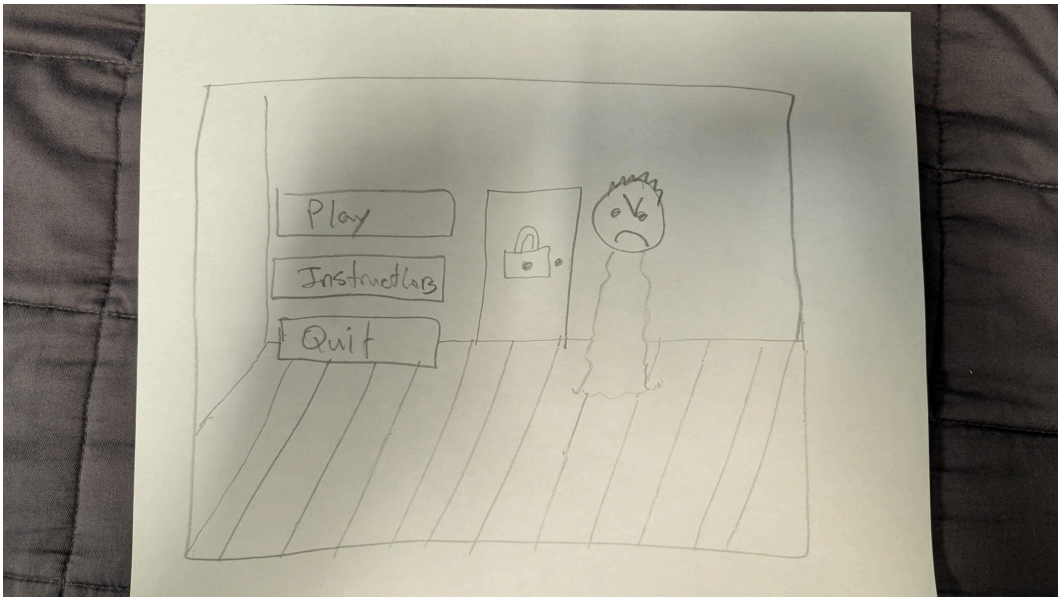
This image shows the player holding a key required to escape



This image shows the player’s first person perspective while holding a flashlight, with the borders of the flashlight’s illuminated area being too dark to see.



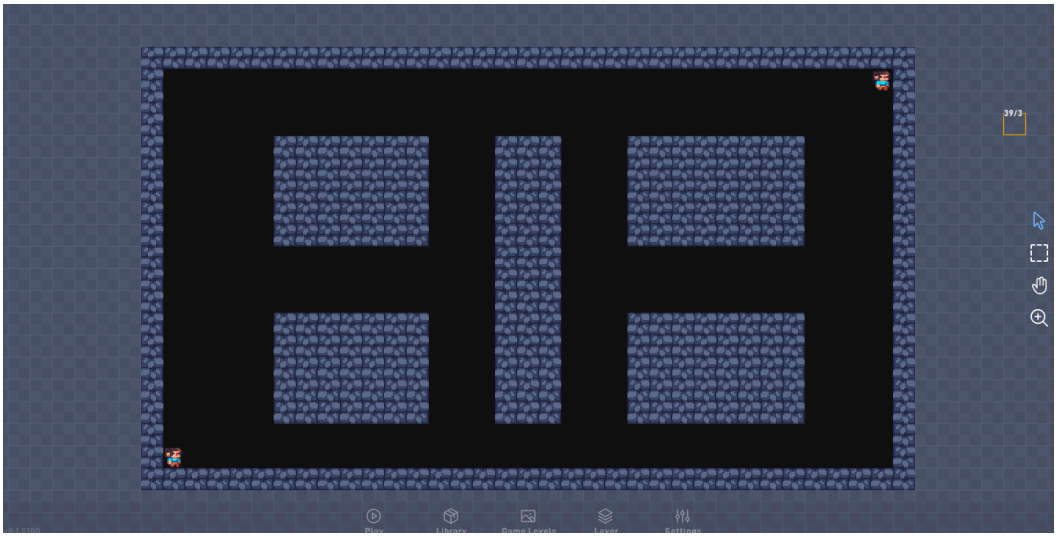
This image shows the main menu.



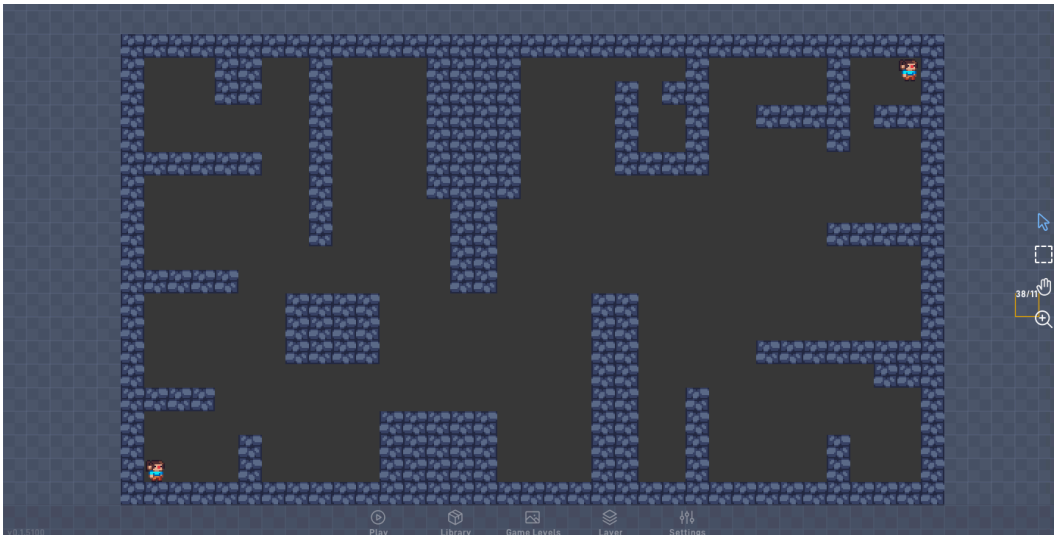
The buttons are designed for relatively simple functions. One of these buttons, the “Instructions” button, will give the player instructions on how to play (eg, what buttons to press, how to move, etc), another “Play” button will start the game proper, and the “Quit” button will quit the game. Although a relatively simple design, putting something like this on paper before going into the much more difficult to manage 3d environment of Unity gave a design that was fixed. Not in the sense that it could not be changed at a later date, but in the sense that when it came time to design the main menu, it would not have to be designed while working within unity, trying to conform objects to fit into a camera and seeing what I could pull off. It was already designed, and simply had to be made to fit this design. The difference might be more properly understood if you imagined yourself trying to figure out where wooden beams are supposed to go while you are in the process of building your house, compared to having a completed blueprint to follow beforehand, then simply implementing it when the time comes. The latter is the far easier and most often used of the two options because it allows you to see things that your mind’s eye may miss, while also giving you a more concrete idea of what you will build so you do not have to design and build at the same time. Designing and building at the same time is much harder than doing both separately.

Another example of how we used paper prototyping is when we designed the levels of the map. Here are a few examples:

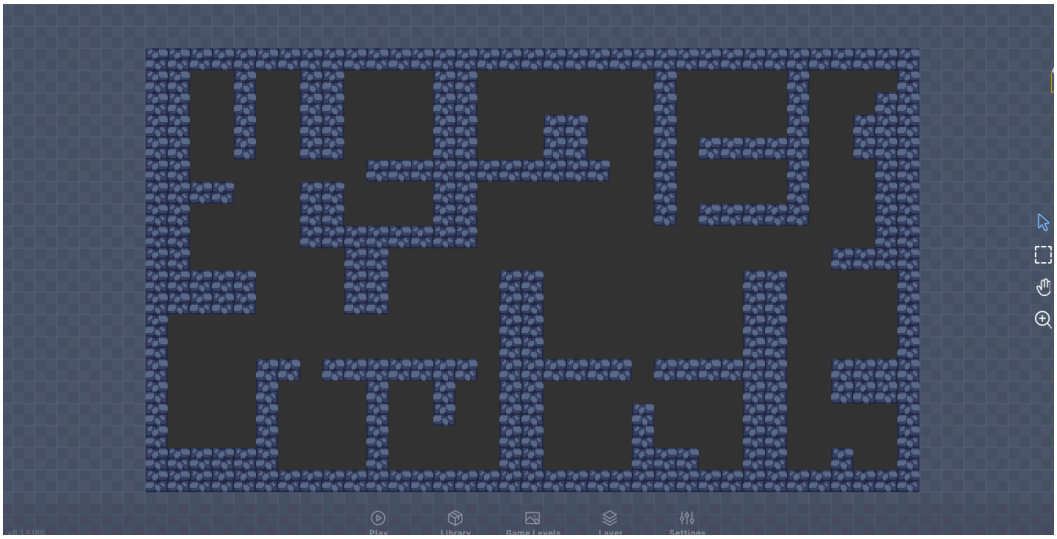
LEVEL 1



LEVEL 2


38/11
Play
Library
Game Levels
Layer
Settings

LEVEL 3


Play
Library
Game Levels
Layer
Settings

Player & enemy movement was modeled simultaneously by printing off these images, then using two coins to represent both the player and the enemy. I would imagine that the player dropped an item in a particular location and, moving both the player both one “square” at a time taking turns, see where the enemy could be that would guarantee a game over. Only the first level idea was implemented in any real way, as it became clear as development occurred that with the enemy being able to hear when you drop an item, dead ends would lead to unavoidable game overs. This forced the design of the later levels to have looping incorporated into the design, so that the player would never truly be cornered due to bad luck. The existence of these paper prototypes was still very useful, as it gave much of the foundation for what would become the actually implemented levels. Through these paper prototypes, much design insight was gained, and the maps were able to be made to both fit design aspects like the ghost following the player and going towards sounds, while also ensuring that the maps were not radically unfair, or force the player into a game over due to back luck through things they could now know or reasonably suspect.

5 GAME TESTING

The testing process for the game was done mainly through developer testing, although we did employ user testing once the game was in a state where there was enough playable content to actually be understandable and functional to somebody who was not on the development team. For the developer testing (which is formally known as “alpha testing”), what was done was essentially an iterative test-driven development cycle. A feature would be developed, code changed, or some other aspect modified to be different from what it was before, then the game would be run in the Unity preview system to see if it acted how it was expected to behave. For example, the item pickup system. This was relatively code-heavy and required a decent amount of scripting to get right. So once changes were made to the script to modify some sort of logic about handling when the interaction button is pressed, how far the player can be from the object and still interact with it, where the item would be floating relative to the player and what rotation it would have, etc, the game would be reloaded and changes tested instantly. This allowed for errors to be corrected very quickly, and for bugs to be minimal. Early on after a change was made, it would immediately be testing quite thoroughly for unintended interactions in order to ensure that previous functionality had not changed when it was not meant to, as well as new features working as intended. For the actual flow of the game to an average player, this was the quick questionnaire that was given.

Are the controls intuitive

Is the game too difficult or too easy, or just right?

Is the enemy too difficult to escape from

Are the puzzles easy to understand

Is it obvious that there is more than one ending

How are the visuals

To the first question, every single person testing the game said the controls felt intuitive. This was confirmed when they only needed, at most, a single reminder in order to remember what button they needed for a specific action. For the game difficulty, this varied greatly. In general, this seemed to be based on how familiar a player was with playing a game in a 3d environment with a mouse and keyboard. One tester did not think to look around as they walked, and so got caught often by the ghost. However, when asked why they were not checking for the ghost before going somewhere, they simply said that they did not play a lot of horror games, and so were not used to doing stuff like that. However, most players were able to quickly understand to avoid the ghost, and the red glow it would give off before turning a corner. When asked, they said the difficulty of the game seemed fair to them, which is interpreted to be that it is not too easy, but not frustrating. For the third question, this varied as well. Some players quickly caught on that tight corner turns were key to escaping, while others simply prioritized not getting seen at all costs, taking abnormally high amounts of time before making moves only when they were sure the coast was clear. These two different playstyles were unexpected, but seem to be the natural result of different players' natural inclinations. Again, only players who self-reported as being unfamiliar with this type of game (specifically the idea of having to run away from an enemy like this) mentioned that it was difficult to escape. Given that this game is intended to have some challenge, this is to be expected. A difficulty option for inexperienced players is certainly a good way to alleviate these problems. For the puzzles, every player reported that the clue system, where papers give numbers and then either an up or down arrow, was obviously connected to the labeled levers with the same numbers. Not a single person was confused on this point, which I believe demonstrates that the game's clue system to this particular puzzle does an excellent job of pointing towards what the player must do without explicit instructions that may impede immersion. When asked if it was obvious if there was more than one ending, the players who completed the game on the normal ending said yes, while the players who did not complete the game by the time they were done testing did not. This is expected, as the clues that there is more than one ending are only really available near the end. Although explicitly stating that there is more than one ending very early on may be a bad idea and take players out of the experience, the game may benefit from some sort of additional hint early on, even if it is very subtle.

For issues brought up by the playtesters, some felt that the game was too dark. However, others said that the darkness went well to make the game scarier. This difference in opinion means that the game may simply appeal to a certain type of player. However, it may be possible that adding some sort of mode where the game is a bit brighter might be good for accessibility purposes, and bringing in more players who might otherwise be interested in the game. The other main issue that some players

brought up was the ghost being too hard. This was exclusively players who self-reported as unfamiliar with games of this type, so the solution may simply be to add an easier difficulty to accommodate these kinds of players in the future. While the game is built with challenge in mind, an easier difficulty can be added separately from the main one without impacting the experience of players who prefer the game as it is normally.

There was also one play session that was recorded, and notes were taken during the session. The player was decently familiar with these sorts of games, and a particular interest was taken in seeing if they would uncover any unintended mechanics that had gone unnoticed, but an experienced player might recognize and be able to take advantage of. Unfortunately, the tester's voice recording did not work, and only the audio of the game itself was recorded. This was not realised until some time later, and by that point it was too late to do another session. Apologies for this oversight. This is the link to the video:

https://youtu.be/rxaJQnn_AHA

This playtest was very insightful as to how an experienced horror-game player behaves. While initially taking a minute to get used to the controls and environment, they very quickly understand how to play, and even take advantage of the "dropping items to lure the ghost" mechanic.

6 GAME BALANCING

An important aspect to any game, especially a game that seeks to offer a challenge, is game balancing. If a game is too easy, people will get bored quickly. If a game is too hard, players will get frustrated and decide it is not worth their time. However, if a game is balanced properly, players will instead feel challenged, and want to engage with the game in order to overcome it. The main way that difficulty was balanced in this game was through the player and enemy speed values. There are no weapons or abilities to speak of in this game, and so they cannot exactly be balanced. However, I believe I have good reason to believe that speed balancing is the key in this situation. If the enemy is faster, they are harder to evade. If they run to a noise faster, you have less time to leave the area after you make a noise. If they are faster when they see you, you have to turn a corner quickly in order to escape. If they are faster when patrolling, then you have less time to do what you want to do in an area before they come back and catch you. The player's speed is set at 15 (this being 15 units per second in the unity ending). This is constant (except while crouching where the player's speed drops), as there is no sprinting system for a temporary speed boost.

The enemy's speeds are not constant. They actually change a lot based on what they are currently doing and what state they are in. This is not explained to the player directly to keep a sense of tension and uncertainty, but is in effect nonetheless. Below is

a table of the enemy's speeds, keeping in mind that the player's speed is essentially constant, since they're not going to crouch in the middle of a chase unless they're doing it to hide in a safe zone.

Enemy State	**Enemy Speed**
Patrolling (Default)	7
Hears Noise	12
Tracking Player	12
Sees Player	16

Player State	**Player Speed**
Walking (Default)	15
Crouching	3

As you can see, the player is faster than the enemy in almost all circumstances. Unless they are crouched (which they normally will never do in the middle of a chase), the player is almost always faster than the ghost. There is one exception though. When the enemy has direct line of sight with the player, and the player is within their vision range, they are 1 unit per second faster than the player. This means that if the player stays in the enemy's line of sight, there is nothing they can do to avoid getting caught. There are areas only reachable by crouching that the enemy cannot follow you into, but the brief moment of time where you drop to a staggering 3 units per second (20% of default) make actually getting into those spaces extremely difficult to time if you are in a

chase. If the enemy is too close to you, you won't have the time. However, they will work if the enemy is decently far away, and you can avoid getting cornered. Now. The main way to balance this game would be to change the speed of either the player or the enemy, and the enemy makes more sense, since a variable player and variable enemy speed would quickly become a mess. Here is what would be a good system for game balance. A difficult system. This was actually intended to be implemented originally, but was cut due to time constraints. Not because implementing it would be too hard, it would only be a few lines of code to change the enemy's speed values based on a difficulty variable changing. It was because balancing it would be a problem. In the current version of the game, the enemy is slightly faster than the player, meaning they need to break line of sight to avoid losing a chase. This is not super difficult, and most of the time it can be done unless you're suddenly cornered due to poor enemy positional awareness. However, what if the enemy speed was higher when they had a direct line of sight, say 20? This was attempted during testing.

This would make the enemy monstrously faster than the player, and during testing, basically made escaping a chase near impossible. It put a massive emphasis on avoiding being caught altogether, because being caught was essentially a death sentence. This didn't make the game unfair necessarily, just way more challenging than intended. The opposite was also true. If the speed of the enemy was lower, say 12, chases were a joke. You were never in danger as long as you kept moving.

The enemy still knows exactly where you are for about 5 seconds after they lose line of sight, but will be slower until they regain it (down to 12 instead of 16). This time of them being slower allows you to escape a chase. Another difficulty modifier would be modifying this speed to be higher or lower depending on whether you wanted the game to be easier or harder, making chases either much easier or much more difficult to break.

Another speed is the speed the enemy movies at when going towards a sound. Currently it is 12. Again, if you made this higher or lower, you would be changing the amount of time you would have after you dropped an item before the enemy showed up looking for you. This increase/decrease in the amount of time you would have to either hide or run away would also impact the difficulty.

The final aspect is the patrolling speed. It sits at a very low 7. This means that if the enemy does not see you or hear anything, they patrol around the room at a very low speed relative to the player. This allows the player to easily evade them if they know where they are and where they are going, provided they don't alert them to where they are. Increasing this by too much would make uninterrupted exploration impossible, so I believe it should not be touched at all, or if it is, then only on a difficulty explicitly meant to be an extreme challenge.

To modify the difficulty of this game, I would say that modifying these speed values to suit different difficulties would be the best way to offer players a version of the game they

can handle. I believe the current speeds are a good default, so they could remain as “normal” mode. An “easy” difficulty would have the enemy’s chase speed reduced to 14 units. Not trivial ,but still slower than the player. A “hard” mode would have it increased to 18, as well as increase the speed at which the enemy heads towards sounds. I believe the lowered enemy speed when they lose line of sight should be unaffected, otherwise escaping a chase may become borderlines impossible. Below are the tables that I believe would best fit the different enemy movement speeds for different implementations of difficulty.

EASY MODE

Enemy State	**Enemy Speed**
Patrolling (Default)	7
Hears Noise	12
Tracking Player	12
Sees Player	14

NORMAL MODE

Enemy State	**Enemy Speed**
Patrolling (Default)	7
Hears Noise	12
Tracking Player	12
Sees Player	16

HARD MODE

Enemy State	**Enemy Speed**
Patrolling (Default)	7
Hears Noise	15
Tracking Player	12
Sees Player	18

EXTREME MODE

Enemy State	**Enemy Speed**
Patrolling (Default)	10
Hears Noise	18
Tracking Player	13
Sees Player	20

The speeds in extreme mode are specifically designed to make it almost impossible to win. However, I have completed the game in development testing with these exact speeds. It was not easy at all, and a lot of patience and strategy had to be developed to do it. The enemy seeing you (unless you were in or right next to a safe zone) was a death sentence. You essentially could not escape. The game shifted from evading a chase to avoiding being seen altogether. The enemy patrolled more quickly, so I had less time to do things after he left the area. Dropping items had to be done at strategic points after calculating which direction the enemy would come from so I could avoid being seen. I believe that a challenge like this would prove interesting to those players who would want a challenge, as it is very hard even for me, and I designed these systems and know the ins and outs of how they work. However, some players like a challenge, and it is completely optional.

I believe this system is the best way to balance difficulty around a player's expectations. If they want a more laid back experience, they can pick easy mode. If they want the intended experience, normal mode. If they want a bit of a challenge, hard mode. And if they want an extreme challenge that even the developer struggles to beat, extreme mode. This way more users can enjoy the game, while the integrity of the intended experience stays intact.

7 PLAYER GUIDANCE

There are two main types of player guidance. The first is direct guidance, and the second is indirect guidance. Direct guidance is where the player is explicitly told to do something (eg, with arrows, something on the HUG, text instructions, a voice telling you what to do, etc), while indirect guidance is something that tells the player what to do, but not explicitly. For example, if an RPG had a very strong enemy that one shot newer players in an area where they aren't supposed to go yet while still technically letting them go there, that is a way of indirectly saying "come back later". Indirect guidance is generally preferable to direct guidance where possible, as too much direct guidance can make the player feel like they are being railroaded (feel like they don't have any freedom to make choices). For this reason, and especially because of the genre that was chosen, almost all of the guidance in the game is indirect guidance. However, there is also a bit of direct guidance that was used.

For direct guidance, there are very few. The first is on the menu. When you click on the instructions button, the player is informed what button to press to interact with and pick up items, what button to press to drop items, what button to crouch, how to move, etc. This form of direct guidance seems necessary, as it is not immediately obvious that the E button is used to pick up items. Things like WASD for movement may be obvious to some players, but not everyone. Additionally, there is a warning that the enemy can hear when you drop items. This is not strictly necessary, as players will learn this fairly

quickly once they actually play the game, but it felt good to warn them about this in case they did not figure out that the ghost hears noises, and instead got frustrated wondering why the enemy kept magically knowing exactly where they were.

The second bit of direct guidance is a note at the start of the game. It is on top of one of the only drawers in the first room, which is essentially impossible to miss. It states that the player must find keys and solve puzzles to escape. Again, the plate may well have figured this out on their own, but if they were confused as to why the door in the second room didn't open, this may help them connect the dots. After this though, practically all of the guidance is indirect.

The third is when the secret ending begins to be triggered. The player must go to the first room again, and so text will appear on the screen saying "I heard something moving in the first room". This is because it is not intuitive to go back there, as no other situation will give you anything from going back to that room.

One form of indirect guidance is the colors of the keys and locks. It isn't explicitly said that the yellow key only works on the yellow lock, or that the blue keys only work on the blue lock, but I have faith that players can figure this one out fairly quickly. There may be issues with colorblind players, and so a colorblind accessibility option may be made in the near future to compensate for this potential issue.

Another form of indirect guidance is the clues to the puzzles. Room 2, for example, has 4 levers, labeled from 1 through 4. The clues you pick up will have a number, as well as an arrow facing either up or down. It is not explicitly stated that these correspond to the levers, but players will very quickly figure this out on their own.

Another form of indirect guidance is the clues for the secret ending. There is a single easy to find blue clue stating, among other things, to "follow my instructions". These instructions are not on this clue, but this leads the player to believe that there are other instructions somewhere without being explicitly told this. The two other notes (which are hidden much better than the ones for the normal ending) have lever instructions. If the levers are pulled according to these notes, the player will be directly prompted to go back to the first room. There, they will see two things that look like bookstands, which will have numbers on them. The player will then realise that the various books they have seen around the asylum actually have a use, and will then go and grab them to put on the stands. After this there will be a purple lock. The player will have dealt with enough of these colored locks to know they need to find a purple key. After this, they find a miniature version of the ghost as a doll, with a note saying "complete the ritual". Again, while not explicitly stated, it is heavily implied that the player must place this doll onto the ritual circle in the third room, as nothing else so far has been able to interact with it.

All of these indirect methods I feel work well to allow the player to feel like they are discovering the solutions themselves instead of being handed every instruction on a silver platter. It allows a sense of fulfillment that wouldn't be possible if every instruction was given on a checklist, or a note just saying every step in order.

Sequencing is another important part of getting a new player to understand your game. If you introduce too much too quickly, the player may be overwhelmed with information. The mechanics of this game are introduced room by room, to make sure the player understands what is required of them before adding a new concept/mechanic.

In the first room, the player must simply open a lock with a key. If they try to open the door with a key, text appears representing the player's internal thoughts saying "I need a key". When they find the key and interact with the door, it opens. All this while avoiding the ghost. Very simple.

The next room is a bit more complicated. You need a key again, but this lock is a different color, and the last key will not work. Instead, you must find a blue key to match the blue lock. There is a purple key as well with no immediately obvious purpose, but it won't work either. This teaches the player very quickly that the color of the lock must match the color of the key. This was true in the last room as well, but there was only one key, so no need to think about it. This is a very intuitive thing that the player will understand almost immediately. However, the next part is more tricky. There is a puzzle that must be completed too. If the payer gets the right key and tries to open the door, it will say "something is still holding it shut". The player must find the 2 clues in the room indicating which way to turn the levers. Once this is done, the door will be able to be opened by the correct key. This room introduces the concept of needing to complete a puzzle to make progress, as well as needing a correctly colored key (which is almost immediately obvious).

The third room is largely the same. 4 levers, but this time marked 5 through 8. There are a few bookstands with marked books as well, though they are seemingly useless. There is also a ritual circle with an altar, although interacting with it will tell the player that "something might go here" but not what exactly goes there. If you do the same thing as in the last room (solve the lever puzzle, find the key, and go through the door), you will get the normal (bad) ending. But this is still an ending that you are meant to get. If the player understands the last room, they can "beat" the game no problem. However, there is a new mechanic that must be understood to reach the secret ending. Backtracking. Once the player has found one of the secret blue clues, they will receive instructions to not only flip a lever in the current room, but also to flip a lever in the previous room. After finding both of them and flipping the levers, a noise will play, and the player will think (in text) "I think I heard something move in the first room". Upon checking the first room, the player will encounter bookstands with numbers on them that weren't there before. This then tells the player that those seemingly useless books from before actually have a purpose. After bringing the books with the corresponding numbers

to the stands, a false wall reveals a purple padlock. The plate already knows what that means. After bringing the purple key, the player gets a voodoo doll of the ghost with a note saying “complete the ritual”. At this point it is obvious what must be done. The player brings the doll back to the ritual circle, and gets the secret ending.

The direct and indirect guidance the plate receives, as well as the slow introduction of new mechanics one at a time, allows the player to properly understand how to beat the game without needing to consult a guide, but also without being explicitly given every instruction. This creates a player experience where I believe they will be able to enjoy figuring things out, without becoming too lost at any particular point.

8 GAME DISTRIBUTION

Game distribution is an essential part of any development project. What platform will a game be on, who will sell it, what hardware will it use, etc. It would be relatively easy to publish a game like this as-is with a distributor like Steam for example. However, a VR release would be a bit more complicated.

In the case of VR, the game would actually require minimal adaptations in order to work. Essentially, the intractable objects in the world would need to have their scripts modified to work with a VR input system. So instead of looking for the E button to be pressed while the player is looking at it, instead it would look for the player to point their VR hand at it, or maybe intersect their hand with the item, then press down on the “grip” part of the controller. This would be used for picking up items, opening drawers, doors, etc. Additionally, it would likely be required that the user keeps the “grip” part of the hand held down while holding items, and releasing it would drop the item. To maintain a proper game balance, likely only one hand would be able to pick up items, so both hands could

not pick up one item each and make the game significantly easier. This would likely be decided by some accessibility option in the main menu where a player chooses which hand is their dominant hand.

Movement would be virtually unchanged. The player would move with a controller stick (which is what VR handheld things have on them for player movement currently), at the same speed of normal gameplay on whatever difficulty they chose. Crouching could either be done with a, or require the player to get much lower in real life. Likely the former, as asking players to crawl in real life to get through an in-game tunnel is a little much, and many players might be unable.

Player/enemy line of sight would have to be tweaked. Instead of the player just having coordinates based on where they are standing, and if those coordinates are in the enemy's line of sight they can see them, it would have to be more dynamic. In VR, the player has a head and two hands. It would likely be changed so that if the enemy could see even one of these things, they could see them. So if you put your hand out from behind a wall and the ghost saw it, a chase would begin. If you peaked your head off from behind a corner while the rest of your body was behind it, chase begins. This may make the game slightly more difficult, but would also make it much more immersive.

Audio elements of the game would remain largely the same. Sound effects play when a door opens, items drop, etc. Music pays throughout the game. Although, it may make the game much more immersive to add a focus on positional audio, to try and allow the player to hear exactly where something is happening. A sound effect of a key dropping playing in both ears when the player just dropped it on their right may be somewhat immersion breaking to more detail-sensitive players.

Something else to consider would be a mobile port of the game. With some small tweaks to reduce graphical quality (file compression, reduced model quality, etc) so that it could run on a smartphone, as well as some coding changes to accommodate the different controls, it may be viable to have a mobile port of the game. This would not only open up a whole new market for sales, but also increase the game's popularity by bringing it to a much younger audience who is more interested in horror themed games than many older consumers.

To generate revenue for this game, the strategy would depend on what platform it released on. If I do decide to release this game on steam, it would likely be a simple 1-3$ price tag, then that's it. However, if the game was released on mobile, I would use a different strategy. Instead of charging for the game, I would make it free, but with advertisements in between player deaths. This is because on PC, people are generally willing to drop a dollar or two on some random game to see what it is all about. However,

on mobile, things are different. The market there is largely children. Even if they have a dollar, they still need to convince their parents to let them buy it, which is a massive barrier to entry. Instead, it would likely be a better idea both for publicity and for revenue, to make the game free on mobile, then have advertisements occasionally.

For any sort of additional revenue, all I can really see are some sort of cosmetics that would cost a small bit of money, like 50 cents to a dollar. Something like making the asylum building Christmas themed and giving the ghost a santa hat could be 50 cents each. Or maybe leaning more into the horror aspect and making a Halloween cosmetic pack that made the asylum look even creepier, and make the ghost even more scary. Things like this would be so inexpensive that it would hardly be an inconvenience, which may lead to more buyers and a larger profit than if they were sold for a higher price with very few buyers. In addition, the mobile port could have something like “pay 1$ to get rid of ads” so that anybody who wanted to pay for the normal experience without interruptions, could get that.

In the event that the game was a success, there are a few more things that could be done. Number one could be updates to the game to get more people interested. If the game gets very popular, add a new area to the map for players to explore, maybe a new type of puzzle, and some more elements to the story. This could get existing players more engaged with the game, while getting potential players interested in being part of an emerging game and its story. There is also the possibility of a sequel game. Maybe in a new location, somehow the ghost returns in an all new map, and the player must once again escape from another haunted building. This could be something like a hospital, a school, a prison, anything really. A whole series of “Escape the Haunted _____”. There could also be something like a prequel to the game, where you play as the person who left the secret notes in the first place. Players could learn the story of how he figured out how to defeat the ghost, how the voodoo doll got to where it is in the first place, and how the trap was set up to defeat the ghost in the main game. While this is very unlikely compared to the spinoff titles, it is something worth considering.

I think that publishing a game like this would not be too difficult, and porting it to new hardware, even something as different as VR, would not be as difficult as one might assume. However, challenges of these kinds are often not fully understood until undertaken. It is likely that there are many aspects of this that would be challenging to deal with that I am simply unfamiliar with. But with enough stubbornness, who knows? Maybe one day this potential game series will be on the front page of the Meta Quest store.

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